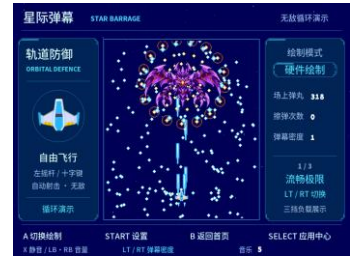
Custom RISC-V 2D Graphics SoC

105 MHz | RV32I + Zmmul

AX7020 / XC7Z020CLG400-2

Native 800 x 600 RGB332



Star Barrage: board capture

AMD

AX7020 / PYNQ / USB / HDMI

INTRODUCTION

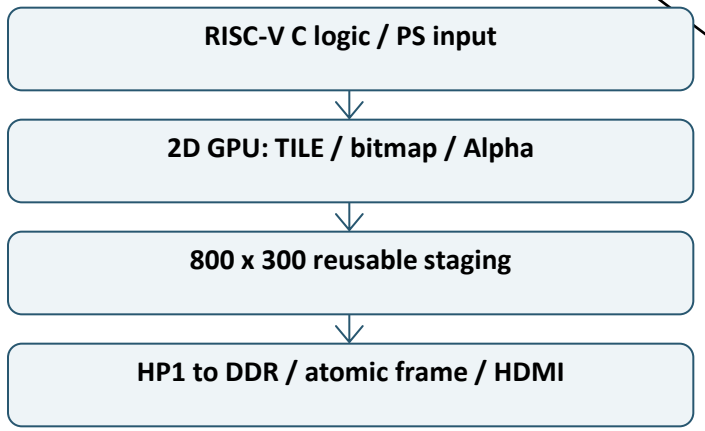
C applications run on a custom 105 MHz RISC-V core with hardware 2D drawing.

One 262144-byte buffer renders two bands; DDR publishes complete 800 x 600 frames.

Touch double-tap and mouse entry pass replay tests. Controller-first USB recovery is verified on the tested devices. LT/RT select three showcase tiers; ARM manages input, PCM and memory.

Main menu: board framebuffer capture





Graphics data path

CPU and system design

Five main pipeline stages and a restricted aligned dual-instruction path.

64 KiB read cache + 4 KiB instruction buffer.

Checksummed retirement before HOLD; keep buffers until DMA drains.

Reproduction and evidence

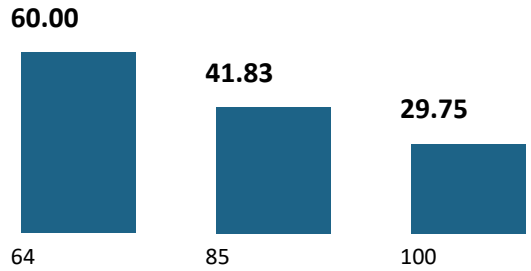
Eight RV32 programs rebuilt byte-for-byte.

Full source-only Vivado rebuild passed.

Reusable artifact checker and PYNQ workflow.

CREATIVE DESIGN

RESULT



Multiple epochs; about 8 s each

Measured results

CoreMark: 371.852 at 105 MHz, 4000 iterations, GCC 13.3 O3/LTO; both CRC runs pass.

TILE 512 / DDR 96 / Alpha 32 and Star density 1: strict 60 for at least 60 s.

System: 23447 LUT, 11576 FF, 126 BRAM36, 4 DSP. CPU hierarchy: 6387 LUT, 2081 FF, 4 DSP.

WNS +0.033 ns; WHS +0.057 ns.

Third-party code/media credited. Physical acceptance and photographs pending.

AMD